SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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27. Do Cropping, Copying and pasting image inside another image using OpenCV.

AIM:
Cropping, Copying and pasting image inside another image using OpenCV.

PROGRAM:

1. **For Cropping**

```
import cv2

img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

if img is None:
    print("Error: Image not found!")
else:
    # Crop the image from (10,10) to (300,300)
    crop_img = img[10:300, 10:300]

    # Save the cropped image
    cv2.imwrite(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\cropped_image.jpg",
crop_img)

    cv2.imshow("Cropped Image", crop_img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

2. **For Copy & Paste**

```
import cv2

# Background image
img1 = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

# Object image (Shinchan)
img2 = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp2.png")

if img1 is None or img2 is None:
    print("Error: One or both images not found!")
else:
```

```python
    # Resize object if it is too large
    rows, cols = img2.shape[:2]

    if rows > img1.shape[0] or cols > img1.shape[1]:
        scale = min(img1.shape[0] / rows, img1.shape[1] / cols) * 0.4
        img2 = cv2.resize(img2, (int(cols * scale), int(rows * scale)))
        rows, cols = img2.shape[:2]

    # Region of Interest
    roi = img1[50:50+rows, 50:50+cols]

    # Convert object image to grayscale
    img2gray = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)

    # Create mask
    _, mask = cv2.threshold(img2gray, 240, 255, cv2.THRESH_BINARY_INV)
    mask_inv = cv2.bitwise_not(mask)

    # Black-out the area in ROI
    img1_bg = cv2.bitwise_and(roi, roi, mask=mask_inv)

    # Extract only the object
    img2_fg = cv2.bitwise_and(img2, img2, mask=mask)

    # Combine
    dst = cv2.add(img1_bg, img2_fg)

    # Place the result back
    img1[50:50+rows, 50:50+cols] = dst

    # Save output
    cv2.imwrite(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\copy_paste_result.jpg",
img1)

    # Display
    cv2.imshow("Result", img1)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:

Image 1:

Image 2:

OUTPUT:

RESULT:

Thus the program is executed successfully.